

Maybank

Nine architectures, two finalists, one decision

A structured tenancy decision for a multi-country CRM deployment across three ASEAN jurisdictions and two business verticals — narrowed from nine candidate architectures to a scored recommendation.

9 → 2

architectures
evaluated

3

jurisdictions

2

business
verticals

13,320

users in scope

7

weighted
criteria

33 v 26

composite
scores

DECISION PARAMETERS · AS SET BY THE PROGRAMME

Three regulators to satisfy simultaneously · one non-negotiable isolation constraint · a five-year infrastructure cost horizon

Aditya Singh · Deployment analysis authored for executive decision and approval · 2026

Both finalists satisfy every regulator and the isolation constraint — so the decision reduces to one variable, and on that variable the consolidated option wins 33 to 26

Situation

A CRM deployment serving 13,320 users across Malaysia, Singapore and Indonesia, split between a consumer financial-services vertical and a wholesale banking vertical.

Three regulators impose distinct data-residency and sovereignty requirements that must all be met simultaneously.

Complication

Nine candidate architectures were evaluated. Most failed on a regulator or on the wholesale isolation constraint.

Two survived — and because both satisfy every hard requirement, the usual compliance argument cannot separate them. The decision turns entirely on the tenancy model for the consumer vertical in the two markets that can share one.

The recommendation

Consolidate the consumer vertical across the two shareable markets into one multi-tenant application with a database per country, and keep the wholesale vertical single-tenant everywhere.

It leads on four of seven criteria, removes one application instance, and preserves identical protection for wholesale data.

THE NUMBERS THAT MATTER

9

architectures evaluated

2

finalists

33 v 26

composite score

4 of 7

criteria won

~25%

fewer app instances

100%

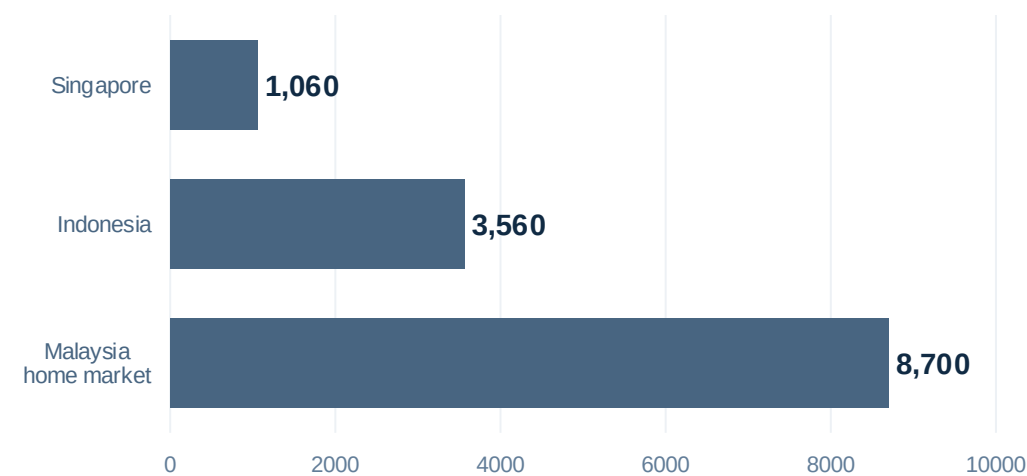
wholesale isolation kept

Why the recommendation is safe to take: the option that wins does not trade compliance for cost. Both finalists were filtered against the same hard constraints first, so the scoring compares only the dimensions where a genuine choice exists.

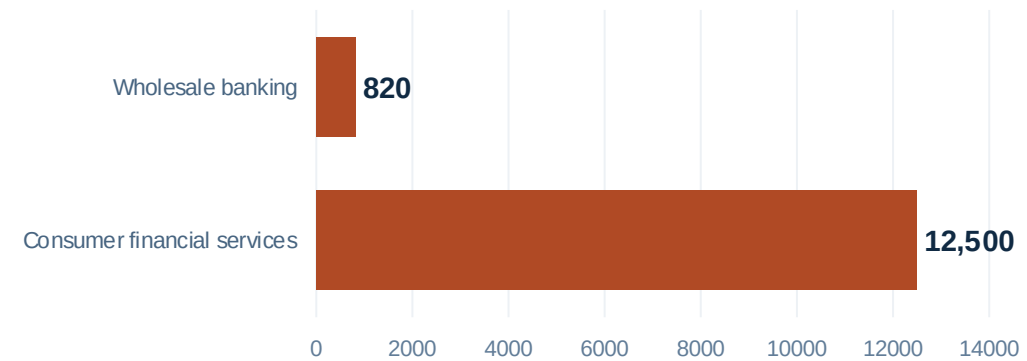
Requirements and constraints were set by the programme; the evaluation framework, scoring and recommendation are mine

SET BY THE BANK Programme context Requirements, constraints and user distribution provided by the programme. Not my determinations. • 13,320 users across 3 markets • Two verticals, distinct governance • Three regulatory regimes • Wholesale isolation as a fixed constraint • Five-year cost horizon • Future-market expansion requirement	ANALYSIS & RECOMMENDATION My contribution The evaluation framework, the narrowing from nine to two, the weighted scoring and the recommendation put to the executive. • Nine-architecture evaluation • Seven-criterion framework • Scoring and composite model • Benefit mapping by option • Risk register and mitigations • Recommendation and sequencing	RANGES, NOT QUOTES Estimated Cost and effort figures are estimates derived from instance counts and operating model, not vendor quotations. • Five-year infrastructure range • Release-effort reduction estimate • Operations headcount effect • Instance-count arithmetic • Sensitive to final sizing • Directionally robust	OUT OF SCOPE Not claimed Outside this deck. It is a decision brief, not an implementation record. • Which option was chosen • Implementation or delivery of either • Any commercial or licence terms • Any confidential bank document • Vendor or platform identification • Post-decision outcomes
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Three regulators, two verticals and a five-year cost horizon meet in a single architectural choice that cannot be revisited after the first go-live



User distribution by market — 13,320 in total



User distribution by vertical — the smaller vertical carries the harder constraint

Regulatory compliance

Three supervisors impose distinct residency and sovereignty requirements. The architecture must satisfy all three at once, not sequentially.

Data isolation

Wholesale client data must be physically separated from consumer retail data. Cross-border customer data must remain ring-fenced per jurisdiction.

Total cost optimisation

Application and database instance counts are the primary cost drivers over five years. Consolidation reduces spend but requires governance guardrails.

Future markets

The architecture must absorb additional markets with minimal incremental infrastructure and onboarding effort.

NARROWING

One fixed constraint eliminated seven of the nine candidates — naming it first is what turned a nine-way debate into a one-variable decision

The non-negotiable

Wholesale banking data is physically separated in every candidate architecture that survived. This is not a scored dimension — it is a fixed constraint reflecting a stricter data-governance posture and specific regulatory obligations for sovereign, government-linked and multinational client data. Any option that logically isolates wholesale rather than physically isolating it was removed before scoring began.

Candidate family	Why it was removed	Failed on
Single regional instance, all markets	Cannot satisfy three residency regimes from one data estate	Residency
Logical isolation for wholesale	Wholesale data co-mingled at storage layer with retail	The non-negotiable
Full multi-tenant, both verticals	Wholesale isolation lost in exchange for cost	The non-negotiable
Offshore-hosted consolidated stack	Indonesian data residency cannot be met offshore	Residency
Per-market, per-vertical, per-segment	Instance count and operational load beyond any cost case	Operability
Hybrid with shared wholesale database	Wholesale databases shared across jurisdictions	The non-negotiable
Consumer split by segment across instances	Fragments the consumer record without a compliance benefit	No benefit

THE FINALISTS

Two architectures survived — identical on wholesale, different only in how the consumer vertical is tenanted in the two shareable markets

OPTION 1
Single tenant, all markets

Every vertical and country pair receives its own dedicated application instance and database.

Home market + regional hub	Consumer: dedicated app, dedicated DB
Home market + regional hub	Wholesale: dedicated app, dedicated DB
Third market	Consumer: dedicated app, dedicated DB
Third market	Wholesale: dedicated app, dedicated DB

4 app instances 4 databases

OPTION 2
Consumer consolidated

Consumer shares one multi-tenant application across the two shareable markets, with a separate database per country. Wholesale stays single-tenant.

Home market + regional hub	Consumer: one app, one DB per country
Home market + regional hub	Wholesale: dedicated app, dedicated DB
Third market	Consumer + wholesale: one app, DB per tenant
Third market	Fully isolated from other markets

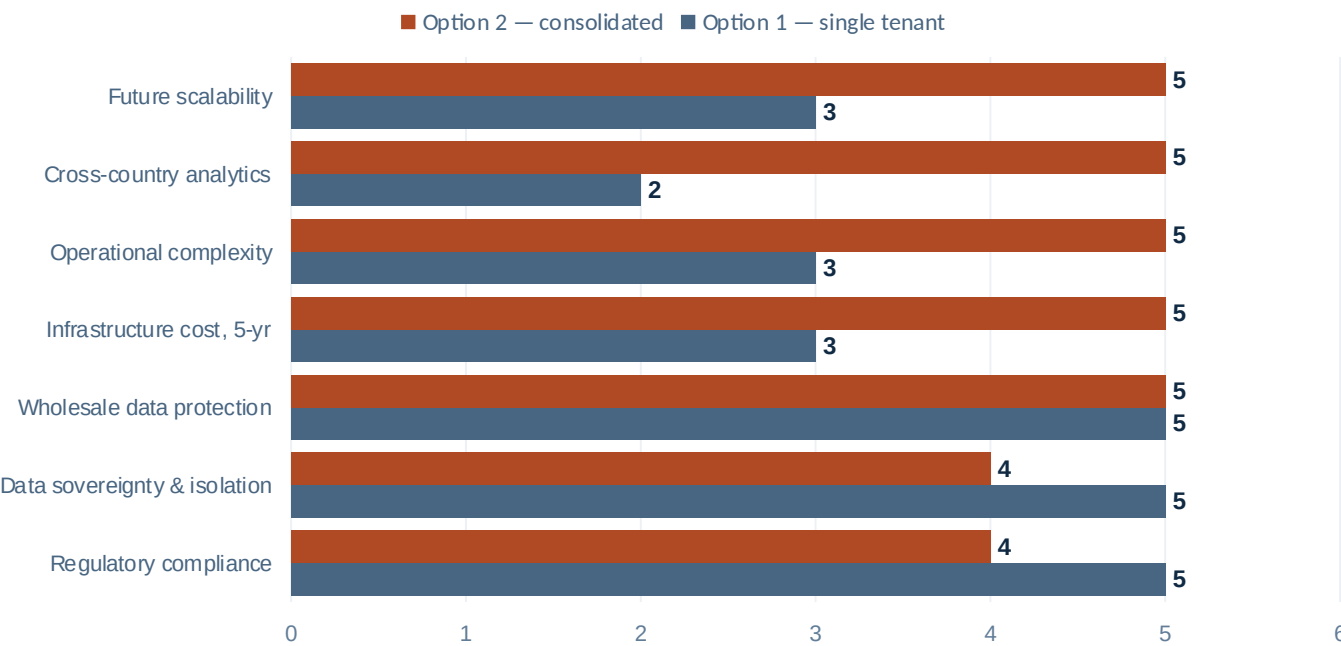
3 app instances 5 databases

The only difference that matters

Wholesale is single-tenant in both options and its protection is identical. The third market is fully isolated in both. The entire decision is whether the consumer vertical in the two shareable markets runs as one multi-tenant application with per-country databases, or as two separate stacks — one fewer application instance, one more database, and a row-level tenancy model doing work that a separate instance would otherwise do.

EVALUATION

Seven criteria, scored one to five — Option 2 leads on four, both are equal on the wholesale dimension, and Option 1 leads only on regulatory simplicity



Scored one to five against criteria and weights agreed with the programme

26

Option 1
composite, max 35

33

Option 2
composite, max 35

Reading the scores honestly

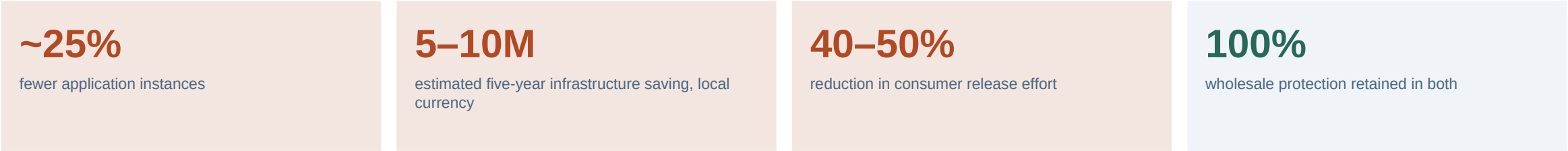
- Option 1 leads on two criteria, both variants of the same point: a simpler compliance narrative for supervisors unfamiliar with row-level tenancy models.
- Both are equal on wholesale protection, so that dimension does not discriminate at all.
- Option 2 leads on the four criteria tied to long-run cost and competitive differentiation.
- The gap is seven points on a thirty-five point scale — decisive, but not so wide that Option 1 is unreasonable.

Where each option genuinely leads

Option 1 buys a simpler regulatory conversation. No multi-tenancy pre-approval is required in either shareable market, and the compliance narrative is one a supervisor can follow without understanding logical isolation. That is a real benefit and it is worth naming.

Option 2 buys everything that compounds: one fewer application instance, unified release management across the two markets, cross-country consumer analytics that the split architecture cannot produce at all, and a pattern that absorbs a fourth market without a new stack.

Consolidation removes one application instance — and the effects compound across infrastructure, release engineering and operations for five years



Business — strategic enablement

- A single consumer customer view across both shareable markets, enabling relationship-led cross-sell for premier and private clients that the split architecture cannot support.
- Product configurations, campaigns and workflows deploy once and apply to both markets — time to market moves from months to weeks.
- Wholesale client confidentiality is unchanged: physically isolated in both options, meeting sovereign and multinational obligations identically.

Technical — platform excellence

- One consumer application means one upgrade event per cycle instead of two, cutting release engineering effort materially.
- Row-level tenancy scopes every query to the correct country entity — a proven enterprise pattern with regional precedent.
- One monitoring stack, one alerting policy and one on-call rotation for the consumer estate; detection and recovery times improve rather than degrade.

Commercial — cost optimisation

- Instance count is the primary infrastructure cost driver; removing one is a durable, compounding saving over the horizon.
- Licence cost is broadly comparable between options — the advantage comes from compute, storage, networking and operations, not per-seat pricing.
- Fewer dedicated administrators and database resources for the consumer estate, with no change on the wholesale side.

The honest framing for an executive: this is not a large saving on a large programme. It is a structural choice whose benefit compounds — one fewer instance now becomes two fewer when the fourth market arrives, and the analytics capability is one the alternative simply cannot deliver at any cost.

RISK

Four risks attach to the recommended option — each with a mitigation that must be committed before approval, not discovered during implementation

Risk	Severity	Why it arises	Mitigation committed before approval
Supervisor unfamiliarity with logical isolation	HIGH	A regulator may require pre-approval for a multi-tenant consumer model, or may simply take longer to be satisfied by it	Pre-engage both supervisors with the row-level isolation design and regional precedent before build starts; hold the fallback to Option 1 open until cleared
Cross-tenant data leakage	HIGH	Row-level tenancy places the isolation boundary in the application layer rather than in physical separation	Tenancy tagging enforced at the data-access layer, not per query; automated cross-tenant tests in the release pipeline; independent penetration test before each market go-live
Shared release train across markets	MEDIUM	One consumer application means one market can no longer release independently of the other	Feature-flag every market-specific change; publish a joint release calendar with agreed emergency-change provisions
Concentration of consumer availability	MEDIUM	A single consumer application is a shared failure domain across two markets rather than one	Multi-availability-zone deployment as designed; documented per-market degradation modes; recovery objectives tested per market, not in aggregate

The first risk is the one that decides the recommendation. If either supervisor will not accept a logically isolated consumer model, Option 2 is not available at any price — which is why regulatory pre-engagement is a gating action before approval rather than an implementation task, and why the Option 1 fallback stays open until it clears.

RECOMMENDATION

Adopt the consolidated consumer architecture, conditional on regulatory pre-engagement clearing in both shareable markets

RECOMMENDED

Option 2 — consumer consolidated

- Leads on four of seven criteria, 33 against 26
- Identical wholesale protection — nothing is traded
- One fewer application instance, compounding with each new market
- The only option that enables cross-country consumer analytics

FALLBACK, HELD OPEN

Option 1 — single tenant everywhere

- Invoked only if either supervisor declines the logical isolation model
- Simpler compliance narrative, higher instance count and cost
- No cross-country consumer analytics capability
- Remains fully viable — this is a fallback, not a failure case

SEQUENCING — WHAT HAPPENS AND IN WHAT ORDER

01

Regulatory pre-engagement

Present the isolation design and regional precedent to both supervisors. Gating action — nothing else starts until this returns.

02

Tenancy design sign-off

Data-access-layer enforcement, cross-tenant test suite and independent penetration test scoped and agreed.

03

Home market build

Largest market first, single tenant in effect until the second joins the shared application.

04

Regional hub onboarding

Second market joins as a tenant. This is the first real test of the consolidated model in production.

05

Third market, isolated

Deployed independently under its own residency regime, unaffected by the tenancy decision.

Four rules for making an irreversible architecture decision defensible

- | | | |
|----|---|--|
| 01 | Filter on constraints before you score | Hard constraints are not scored dimensions. Putting them into a weighted scorecard invites people to trade away something that was never tradeable, and turns a decision into a negotiation. |
| 02 | Reduce the decision to one variable | Nine options is a debate. Once both survivors are identical on every hard requirement, the decision has exactly one variable in it — and an executive can actually take it. |
| 03 | Name what the losing option genuinely wins | Every real option has a real advantage. Stating Option 1's simpler regulatory narrative is what separates a decision brief from advocacy, and it is what makes the recommendation trusted. |
| 04 | Make the deciding risk a gate, not a task | The risk that could invalidate the recommendation belongs before approval, with the fallback held open — not in an implementation plan written after the decision is announced. |

The framework, not the answer, is what transfers. Nine to two to one variable, scored against agreed criteria, with the deciding risk gated before approval — that shape works for any irreversible architecture choice.